Negative Review or Complaint? Exploring Interpretability in Financial Complaints

Sarmistha Das, *Graduate Student Member, IEEE*, Apoorva Singh, Sriparna Saha, *Senior Member, IEEE*, and Alka Maurya

***Abstract*—In the financial service sector, customer service is the most critical tool for long-term business growth. A financial complaint detection (CD) system could aid in the identification of shortcomings in product features and service delivery. This could further ensure faster resolution of customer complaints and thereby help retain existing clients and attract new ones. Prior research has prioritized only complaint identification and prediction of the corresponding severity levels; the first aim is to categorize a textual element as a complaint or a noncompliant. The other attempts to classify complaints into several severity levels based on the degree of risk the complainant is willing to endure. Identifying the reason or source of a complaint in a text is a significant but underexplored area in natural language processing study. We propose an explainable complaint cause identification approach with a dyadic attention mechanism at the sentence and word levels, enabling it to give varying amounts of emphasis to more and less important information. As the first subtask, the model simultaneously trains CD, sentiment detection, and emotion recognition tasks. Afterwards, we identify the complaint's cause and its severity level. To do this, the causal span annotations for complaint tweets are added to an existing financial complaints corpus. The findings suggest that conventional computing techniques can be adapted to solve extremely relevant new problems, generating novel opportunities for research[1].**

***Index Terms*—Complaint severity detection, deep learning, explainability multitask learning, financial complaint corpus, financial complaint detection.**

I. Introduction

COMPLAINING is an expression of negative emotions communicated due to a discrepancy between reality and expectations [1]. Classification of complaints in the realm of pragmatics theory has been established through a nuanced differentiation of levels of complaint severity, and the examination of the extent of prenotification the complainant is inclined to accept, as well as an analysis of the underlying motivation of the complainant [2], [3]. Recent studies on complaint detection (CD) have mainly dealt with automatically identifying binary complaints and the associated severity levels from social media data [4], [5], [6]. However, these studies primarily focused on improving CD performance with the help of various models without providing any evaluation of the outcome's explainability. Since the introduction of explainable artificial intelligence (X-AI) [7], providing interpretation for any AI algorithm's decision has become essential. Therefore, rather than enhancing performance by increasing computational burden, there is a push to construct trustworthy and transparent interpretable models.

Other than popular applications such as chatbot support and task automation, AI in finance could be used to resolve digital service grievances. Retail banks, government institutions, and insurance companies engage with many online consumers and are more likely to be involved in financial disputes [8]. Contrary to traditionally classified complaints, interpretable financial CD modules could enable these organizations to route them to suitable divisions efficiently. This can encourage integrity and accountability when handling financial products and services, thereby streamlining the process and enhancing market presence. The identification of the scope of causality in relation to the emotions conveyed is considered a fundamental advancement in the area of research involving automatic reasoning in human emotions [9], [10]. Using these associated studies as a guide, we intend to determine the cause or reason for the financial complaints posted on the social web. The present complaint datasets are of immense value for the training of CD models; however, they are limited in that they only comprise binary complaint labels and levels of severity. To establish the reliability and practicality of such frameworks, it is crucial to augment them with interpretability and further investigate their workings. Often, inaccurate information leads to the widespread occurrence of fraudulent transactions. Fig. 1 depicts an instance of a severe complaint caused by a fraudulent transaction. The customer unintentionally made the credit card payment and is now seeking assistance. Evidently, the sample tweet is of accusative nature. Therefore, the given example can be identified as a complaint with the severity level of an accusation. The primary cause of the tweet being classified as a complaint is the fraudulent transaction made through the credit card.

1) Research Objectives: Following are the research objectives of the current study.

Manuscript received 4 August 2023; revised 18 October 2023; accepted 16 November 2023. Date of publication 2 January 2024; date of current version 31 May 2024. *(Sarmistha Das and Apoorva Singh are co-first authors.) (Corresponding author: Sarmistha Das.)*

Sarmistha Das, Apoorva Singh, and Sriparna Saha are with the Department of Computer Science and Engineering, Indian Institute of Technology, Patna 801103, India (e-mail: sarmistha_2221cs21@iitp.ac.in).

Alka Maurya is with Crisil Private Ltd., Mumbai 400 076, India (e-mail: alka.maurya_@crisil.com).

Digital Object Identifier 10.1109/TCSS.2023.3338357

[1]The code and dataset are available at https://github.com/sarmistha-D/Complaint-HaN

2329-924X © 2024 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Sivas Bilim ve Teknoloji Universitesi. Downloaded on June 09,2026 at 07:36:48 UTC from IEEE Xplore. Restrictions apply.

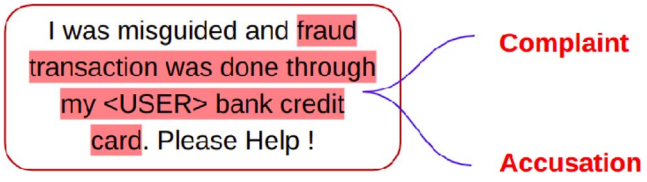


Fig. 1. Example of complain cause analysis. Highlighted portion indicates the causal span for the complaint.

a) Our objective is to ascertain the justifications that lead to the categorization of financial content as complaints.
b) In addition, we aim to understand causal information's effect on the CD and severity classification (SC) tasks in the proposed model.
c) We aim to comprehend the effect of sentiment and emotion recognition (ER) tasks individually and in different combinations in the proposed model.

2) Contributions: Our work's major contributions are as follows.

a) We identify the rationale/cause of social media data classification that pertains to complaints, specifically in the financial domain. To the extent of our knowledge, this study constitutes a novel endeavor in the explainable financial complaint identification field.
b) We performed an extension to a publicly accessible dataset through the addition of manual annotations regarding the causal scope of sentences labeled as either complaints or noncomplaints (NCs).
c) We propose a hierarchical attention (HAtt)-based framework that trains jointly on the following tasks: i) binary classification of complaints; ii) recognition of emotions; iii) categorization of sentiment as the initial subproblem and the following subsequent subproblem entails; iv) extraction of causes; and v) classification of severity levels. The main tasks in our multitask approach are CD, cause extraction (CE), and severity level categorization (SC). Contradictory, ER and sentiment analysis (SA) are considered supporting tasks.

The rest of the article is structured as follows. Section II research describes some of the earlier studies on this subject. We go into dataset generation and annotations in great detail in Section III. In Section IV, we explain our proposed technique for multitasking studies. Section V discusses the experiments we conducted and the results obtained. Finally, in Section VII we summarize our findings and discuss future directions.

II. Related Work

In the subsections below, we discuss the existing research on automatic CD from social media content.

A. Complaint Detection in Pragmatics and Computational Linguistics

Complaining can be used to express disapproval, seek explanations, or both [3]. In pragmatics, a group of complaints can be articulated into severity levels [2] depending on the straightforwardness, emotional intensity, and the intent of the complainant: 1) no specific reproach; 2) disapproval; 3) accusation; and 4) blame.

Previous studies in computational linguistics focused solely on complaint classification using feature-based machine learning methods [4], [12] and deep learning models based on transformer networks [5], [6], [13]. However, a recent development in complaint analysis involves analyzing complaints at the aspect level [14]. This approach provides valuable insights into the specific attributes or aspects that users are complaining about.

B. Complaint Detection in Risk Management

In recent times, complaint mining research has concentrated on commodity hazards and risks [15], along with the likelihood of complaint escalation [16]. Subsequently, an advanced model was developed focusing on the analysis of the severity level of complaints [6] by training multiple transformer models coupled with linguistic information to determine severity levels in complaints.

C. Multitasking in FinTech Services

In various fields, deep-learning-based multitask models have become the preferred choice due to their ability to learn multiple tasks jointly. Subsequently, in 2017, Isonuma et al. [17] developed a multitask setting configured model for financial document classification and extraction. Recently, a complaint analysis model based on financial content has been developed to enhance CD within the financial domain [11]. This model integrates sentiment, emotion, and severity information, aiming to improve the overall accuracy of CD. Additionally, in 2020, Yang et al. [18] demonstrated the prediction of financial volatility using transformer-based multitask configurations to represent price variability on financial assets over a certain period of time.

D. Cause Extraction

Emotion cause extraction (ECE) refers to extracting specific stimuli underlying emotion within a text. Earlier in 2010, Lee et al. [9] first introduced ECE with rule-based techniques. Gradually, in 2021, Poria et al. developed an advanced ECE task in dialogues [10] in order to enhance the explainability of emotion-aware models.

However, there have been no studies on causal span detection and extraction of complaints using computational methods specifically in the financial domain, even though there are several in the related field of ER [10], [19]. Financial data is typically presented in an unstructured format, which leads to frequent noise and ambiguity in textual complaints. Overlapping and ambiguous complaint categories pose challenges in accurately determining complaint class distribution, hindering the ability to extract relevant information effectively. We present an interpretable complaint cause analysis model to address this challenging task, which is based on a HAtt-based network approach at the word and sentence-levels and takes advantage of

Authorized licensed use limited to: Sivas Bilim ve Teknoloji Universitesi. Downloaded on June 09,2026 at 07:36:48 UTC from IEEE Xplore. Restrictions apply.

TABLE I
COMPARATIVE STUDY OF EXISTING COMPLAINTS AND *X-FINCORP* DATASET

Dataset	Labels				
	Complaint	**Severity**	**Sentiment**	**Emotion**	**Cause**
Complaints [4]	✓	×	×	×	×
Severity_based_complaints [6]	✓	✓	×	×	×
Complaints-ESS [5]	✓	✓	✓	✓	×
FINCORP [11]	✓	✓	✓	✓	×
X-FINCORP (current work)	✓	✓	✓	✓	✓

the multitask assign paradigm to varying amounts of emphasis to more and less important information. For this study, we added the causal span of complaints to a corpus of Twitter-based financial complaints.

III. CORPUS DETAILS

Many financial institutions and service providers utilize Twitter accounts for customer support purposes. Due to the platform's microblogging nature, customers find it more convenient to express their complaints and seek assistance on Twitter compared to traditional methods like sending emails or visiting physical locations. This trend is attributed to the ease and speed of communication offered by Twitter accounts maintained by these organizations. This influences us to utilize the *FINCORP* dataset[2] [11], which is a Twitter-based financial complaint dataset. The dataset comprises 3133 typed samples of NCs and 3149 samples of complaint-typed instances in the English language. We selected this publicly available dataset deliberately, as it differs from other existing datasets (such as those mentioned in [4], [5]) that suffer from imbalances and primarily contain generic complaints from various domains. In contrast, our dataset specifically focuses on complaints related to financial organizations and their customers. In addition to the complaint identification axis, this dataset includes three other related aspects: severity levels (covering categories like *no explicit reproach, disapproval, accusation, blame, and NCs*), emotions (comprising *anger, disgust, fear, happiness, sadness, surprise, and other*), and sentiments (encompassing *negative, neutral, and positive*). Figs. 2, 3 and 4 provide a graphical representation of the distribution of data instances for each class label. These figures illustrate how the data is divided among the different severity levels, emotions, and sentiments present in the dataset. The "other" emotion class represents tweets that do not fit within Ekman's six primary emotions [20]. The present study concentrates on broadening the current corpus by manually annotating causal spans for complaint class labeled sentences, allowing for multifaceted research. Table I compares the existing Complaints dataset and the extended *X-FINCORP* dataset.

[2]https://github.com/RohanBh23/FINCORP

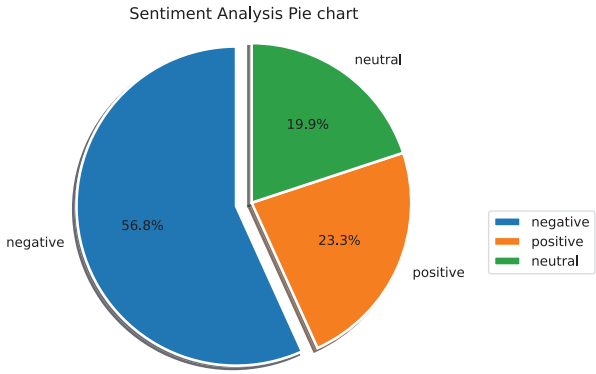


Fig. 2. Distributions of class labels of sentiment classes.

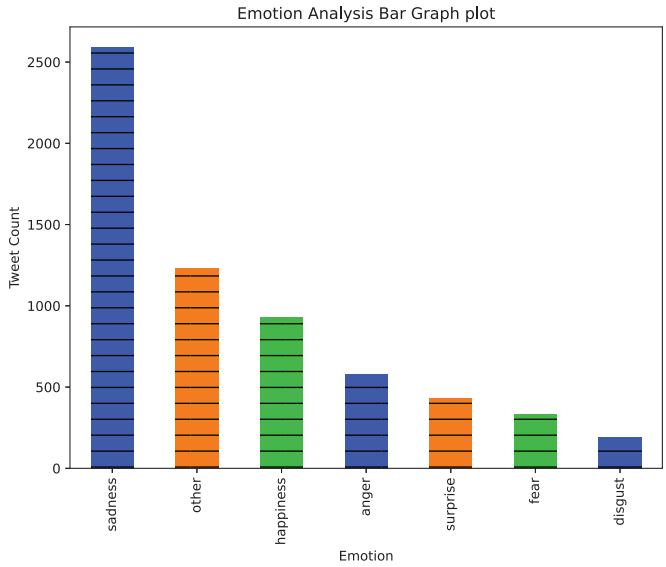


Fig. 3. Example of class labels of emotion classes.

A. Cause Extraction Method

1) Task Definition: Complaint Cause entails a textual component where the user explicitly states the rationale for raising a complaint. It serves as the individual's speech act, specifically employed to communicate the specific conditions or instances in which their expectations have been unmet or violated, leading them to express their grievances. This aspect of the

Authorized licensed use limited to: Sivas Bilim ve Teknoloji Universitesi. Downloaded on June 09,2026 at 07:36:48 UTC from IEEE Xplore. Restrictions apply.

TABLE II
SAMPLE INSTANCES WITH THEIR CAUSE ANNOTATION FROM THE *X-FINCORP* DATASET. LABEL: LABELS FOR COMPLAINT, SENTIMENT, AND EMOTION TASKS

Tweet Text	Cause
< USER > Authorization charge has not been refunded over a month now. Can someone help me ***Label:*** *complaint, sadness, negative*	Authorization charge has not been refunded over a month
I request all ppl do not use < USER > cards as they charge for redemption of reward points. They promote online transaction to avail reward points and then do fraud! ***Label:*** *complaint, anger, negative*	Do not use < USER > cards as they charge for redemption of reward points
< USER > We thank you for approaching the Income Tax Department regarding your grievance. We are looking into your grievance on behalf of Income Tax Department and will revert to you shortly. Inconvenience caused if any, is regretted. ***Label:*** *noncomplaint, happiness, positive*	No cause

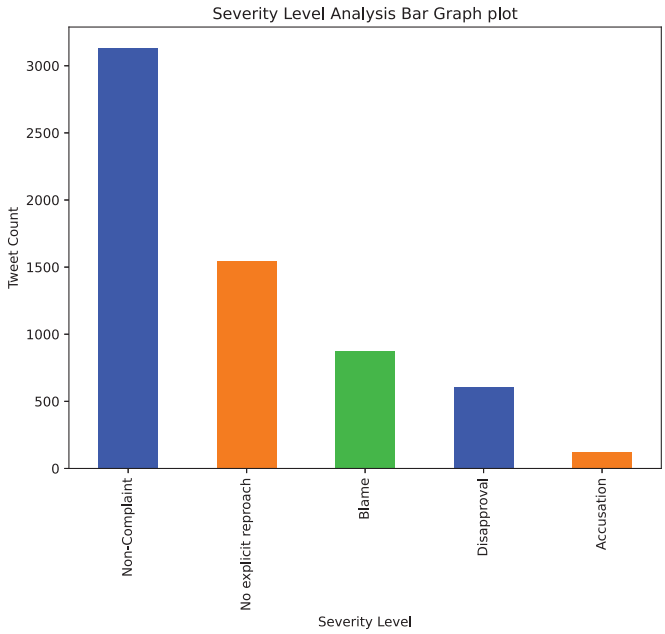


Fig. 4. Example of class label of severity classes.

complaint provides crucial information about the underlying reasons behind the customer's dissatisfaction, enabling a better understanding of the root cause of the issue.

2) Annotations: The sensitive annotation task of causal span identification for each sample was assigned to three annotators: two undergraduate students and one doctoral student from the computer science field. The selection process for undergraduate annotators involved a rigorous two-phase competition, open exclusively to students majoring in computer science and economics disciplines. The doctoral annotator possessed the requisite expertise for constructing supervised, domain-specific annotated corpora. Initially, the expert annotator annotated 282 samples to create the gold standard annotations. In the first phase of the annotation program, 200 samples were distributed, with each participant tasked to annotate 50 samples. Selection for the second phase was contingent on the quality criteria encompassing semantics, soundness, and coherence. Those who met these criteria advanced to the second phase, where they were assigned an additional 80 samples for annotation. Ultimately, only two annotators were chosen based on their ability to fulfill the annotation criteria. Each participant received an honorarium for contributions in each annotation phase. Finally, the two selected undergraduate annotators were assigned the remaining 6000 tweets for annotation. They were instructed to work separately and subsequently verify their annotations through mutual discussion to determine the gold standard summaries. To prevent annotation biases, the annotators were instructed to segregate and identify the span of the text that pertains to the cause S(I) in isolation, that accurately portrayed the fundamental reason behind each occurrence of complaint (C), for instance (I) in the *X-FINCORP* dataset.

For instances categorized as the NC class, the annotators labeled the sentence as *no cause*. We used the macro-F1 [21] metric to measure the interrater agreement, as supported by previous studies on span extraction. The resulting F1 score of 0.81 indicates the cogent quality of annotations[3]. The proposed extended corpus used in this research work contains several financial domains along with the tweet text, complaint label, sentiment and emotion label, severity level, and the corresponding annotated causal span for each record. Table II exhibits several examples of annotated causal spans.

[3]The authors in [11] mentioned the agreement ratings of complaint, severity level, emotion, and sentiment tasks as 0.83, 0.69, 0.68, and 0.82, respectively, suggesting good annotations.

Authorized licensed use limited to: Sivas Bilim ve Teknoloji Universitesi. Downloaded on June 09,2026 at 07:36:48 UTC from IEEE Xplore. Restrictions apply.

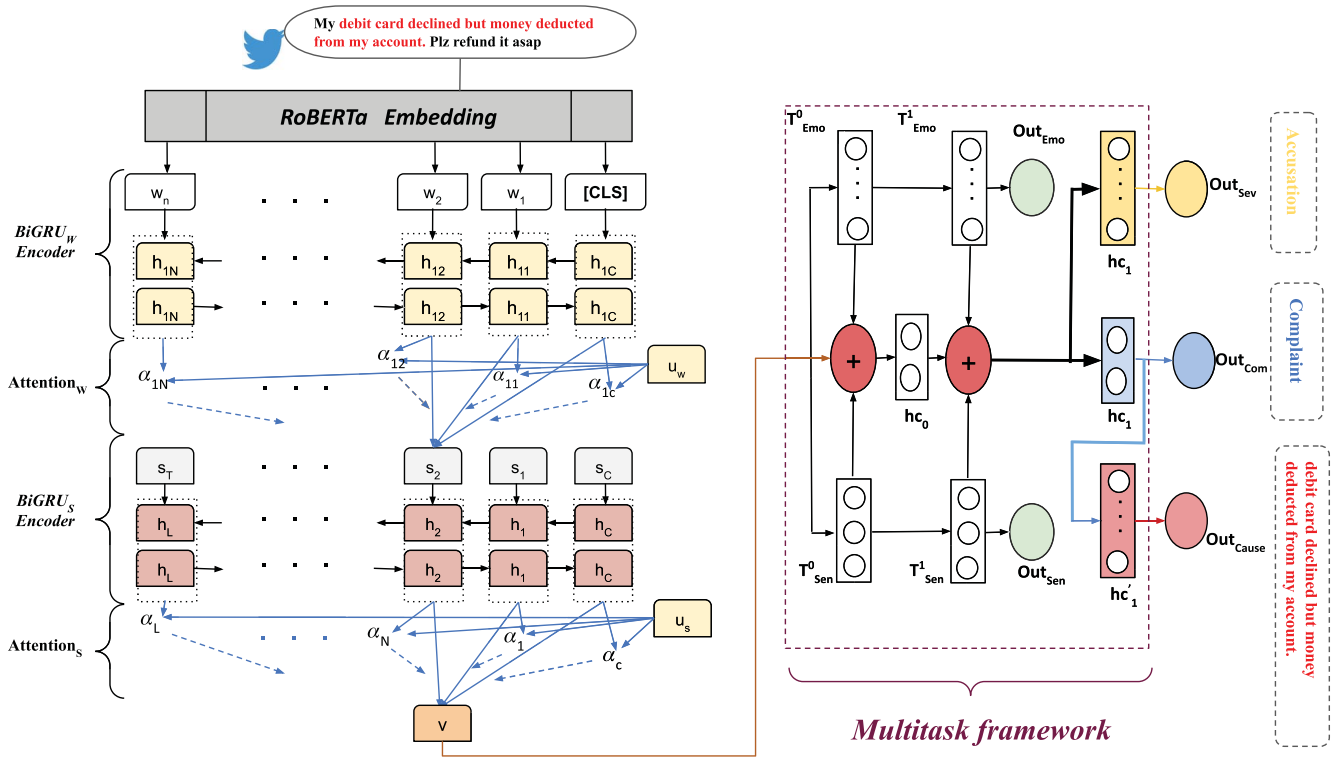


Fig. 5. HAtt-based complaint-cause detection and extraction framework. W, word-level and S, sentence-level.

IV. Methodology

This section discusses the proposed architecture in detail. Fig. 5 depicts the whole framework.

A. Problem Definition

We tackle the CE and related severity level classification as two separate subproblems: 1) complaint identification employing a multitasking framework that treats sentiment and emotion classification as supplementary tasks; and 2) while simultaneously conducting the extraction of causality and predicting the severity of the issue. We define the extraction of the causal span task for any specimen as follows: Given $X_t = \{x_0, x_1, .., x_t, .., x_n\}$ text with complaint class label c, sentiment label s, emotion label em, and severity level m, then determine the causal span $ce(X_t)$ of instance X_t. Here, $X_t \in n$ and n is the number of text instances. complaint labels ($c_t \in C$), sentiment labels ($s_t \in S$), emotion labels ($em_t \in E$), severity labels ($m_t \in Sev$), and CE ($ce(X_t) \in X_t$). Our multitask model jointly learns to maximize prediction that links each instance of X_t with appropriate complaint label (c_t), sentiment label (s_t), emotion label (em_t), severity level (m_t), and CE label ($ce(X_t)$)

$$\text{argmax } \theta \left(\prod_{t=1}^{n} P(c_t, s_t, em_t, m_t, ce(X_t), |X_t, \theta) \right) \quad (1)$$

where θ is the optimized parameter.

B. Textual Features

Bidirectional encoder representations from transformers (BERT) are an encoder that utilizes transformers and is both multilayered and bidirectional. This encoder was introduced in prior literature [22]. It has been applied in previous research endeavors [23]. It is trained in masked language modeling, which requires randomly masking specific tokens in the input to predict them solely based on context. RoBERTa [24], an extension of BERT, has shown improved performance on social media analysis tasks [25] by being trained on a larger data set with varying hyperparameters. For this study, we employed the *RoBERTa-base-uncased* model to represent the words in each tweet instance.

C. Hierarchical Attention (HAtt) and Word Attention Module (WAtt)

To increase the context richness of our data sample representation, we enlarge our model to a hierarchical structure; we compute the sentence representation by utilizing sentence-level encoding, for a given input text collection instances, $\text{XT} = [St_1, St_2, St_3, ..., St_n]$ from the source data, where a given sentence, St_i, is fabricated of word tokens $[\text{WT}_{l-k+1}, \text{WT}_{l-k+2}, \text{WT}_{l-k+3}, ..., \text{WT}_l]$. In this approach, the word tokens from an entire tweet are considered as a unified sequential input and processed by a Bi-GRU encoder, functioning similar to a word-level encoder. At each time step, the output of the input word token, denoted as wL_t, is represented as h_t^{wL}. We use the superscript wL to signify the word-level encoding,

Authorized licensed use limited to: Sivas Bilim ve Teknoloji Universitesi. Downloaded on June 09,2026 at 07:36:48 UTC from IEEE Xplore. Restrictions apply.

while sL is used to indicate the sentence-level encoding. This method enables the encoding of the tweet's textual content into a sequential representation, capturing both word-level and sentence-level information for further analysis and processing.

Word-level attention (WAtt) has been introduced because not all words add the same amount of meaning to a sentence. The combined representation of those attention-based words is used to form a sentence vector as follows:

$$h_{st_i}^{wL} = \sum_{t=1}^{k} a_{l-k+t}^{wL} h_{l-k+t}^{wL}. \tag{2}$$

Thus, we obtain word-level sentence embedding of the instance, $h_T^{wL} = \left[h_{s_1}^{wL}, h_{s_2}^{wL}, h_{s_3}^{wL}, ..., h_{s_n}^{wL}\right]$. Then the embedded sentences will be fed into Bi-GRU as a sentence-level encoder. Subsequently, we obtain final sentence-level embedding $h_T^{sL} = \left[h_{s_1}^{sL}, h_{s_2}^{sL}, h_{s_3}^{sL}, ..., h_{s_n}^{sL}\right]$ which is aggregated with attention to obtain a tweet representation, h_{st}^{TR}, as follows:

$$h_{st}^{TR} = \sum_{j=1}^{n} a_{s_j}^{sL} h_{s_j}^{sL}. \tag{3}$$

We pass these tweet representations into the multitask module to evaluate the effectiveness of complaint, severity, and CE using sentiment and emotion information.

D. Multitask Framework

As part of our study, we configure the CentralNet [26] framework into a multitasking architecture. The CentralNet multitask architecture is a neural network approach where individual task-specific channels are separated into n distinct networks, each with its own central network. The central network is the primary layer for detecting complaint and severity labels as well as CE. It integrates the features obtained from various single tasks (such as sentiment and emotion classification) by analyzing the weighted combination of the outputs from task-specific networks and the layers preceding them. The multitasking layers can be described as follows:

$$\text{MultiL}_{i+1} = \alpha_m \text{MultiL}_i + \sum_{k=1}^{n} \alpha_{s_i^k} \text{HT}_i^k. \tag{4}$$

This model consists of n task-specific networks, where n represents the number of individual tasks and α are defined as scalar weights adjustable during training with α_m designated for the primary task and α_s allocated for the subtask. The hidden representation HT_i^k of each task-specific network k at the ith layer is combined with the central hidden representation MultiL_i of the main task. The combination is used to produce the final layer MultiL_{i+1}, which is then fed into an operating layer, consisting of a dense layer and an activation function.

E. Cause Extraction Postprocessing

The postprocessing for the CE task is executed by utilizing the binary labels (0 or 1) assigned to each word token by the model, where 0 represents NC ($0 => NC$) and 1 represents complaint ($1 =>$ complaint). The labels are then mapped (label $\in \{0, 1\}$) to each word token within the tweet text. From the token length within the sentence, the starting and ending points of the causal span can be derived directly. For example: Please/[0] help/[0], my/[1] IMPS/[1] transaction/[1] number/[1] 136401602410/[1] got/[1] stuck/[1]. please/[0] do/[0] the/[0] needful/[0].

F. Loss Function

1) The categorical-cross-entropy (J_{CE}) losses are computed for the CD, ER, SA, and severity categorization (SC) tasks. The combined loss function ($J(\theta)$) is formulized as follows:

$$J(\theta) = J_{\text{CE}}^{\text{CD}}(\theta) + J_{\text{CE}}^{\text{ER}}(\theta) + J_{\text{CE}}^{\text{SA}}(\theta) + J_{\text{CE}}^{\text{SC}}(\theta). \tag{5}$$

The symbol θ denotes all the model parameters to be optimized.

2) To measure the model's performance for the CE task, we utilize the binary cross-entropy as the loss function. The binary cross-entropy (J_{BE}) allows the model to assess whether a given instance is a member of a specific class or not, even if the instance can belong to more than one class.

3) The combined loss function (L) of our proposed model is

$$L = J(\theta) + J_{\text{BE}}. \tag{6}$$

V. Experiments, Results, and Discussion

The research outcomes of several variants of the proposed framework tested on the *X-FINCORP* dataset are described in this section.

A. Baselines

1) Multitask Systems: We developed Baseline_1 [5] model as a multitasking baseline based on recent advancements in the field of CD within a multitasking framework. Baseline_1 is a multitask deep learning model based on the BERT architecture, designed for complaint, emotion, sentiment, and sarcasm classification. We have modified this model to include joint learning of CD, SC, and CE tasks, with SA and ER as additional tasks. The experimental design remains similar to our current study. At the conclusion of the binary and multiclass prediction process, we implement four individual output layers, each incorporating a softmax activation function to respectively handle the CD, SC, ER, and SA tasks while utilizing a sigmoid function for the CE task.

2) Baselines for Cause Extraction Task: We approached the novel task of CE in the domain of complaint analysis by drawing inspiration from prior work in ER [10]. To this end, we leveraged a pretrained SpanBERT base model, fine-tuned on the SQuAD 2.0 dataset [21], specifically for the purpose of performing the CE task.

Authorized licensed use limited to: Sivas Bilim ve Teknoloji Universitesi. Downloaded on June 09,2026 at 07:36:48 UTC from IEEE Xplore. Restrictions apply.

3) Ablation Models: We have implemented the suggested framework in two configurations: HAtt + CentralNet model and WAtt + CentralNet model. Further, to analyze the individual influence of emotion and sentiment classification tasks on the CD, categorization of severity level (SC), and extraction of cause (CE) tasks, we have devised multiple ablation models while maintaining the CD, SC, and CE tasks fixed. The remaining architectural aspects of these models are similar to the proposed system.

B. Experimental Setup

We executed the implementation of our proposed model and all the baseline models utilizing the PyTorch[4] [27], and Tensorflow[5], both built-in Python programming language, along with the Scikit-learn[6] [28] library. Our experiments were executed on a multiGPU hybrid cluster consisting of RTX 2080Ti GPUs. The training configuration was maintained consistently across all experiments [6]. The models were evaluated utilizing a nested 10-fold cross-validation approach, which involved a double-loop process. In the outer loop, nine-fold cross-validation of the data was utilized for training, and one-fold was used for testing. In the inner loop, the nine-fold cross-validation obtained from the outer loop was further subjected to three-fold cross-validation, with two-fold used for training and one-fold being utilized for validation. In order to determine the optimal model during training, we adopt a strategy of minimizing the validation loss over 30 training epochs. The input sequence is set to the maximum length of 20. We use a batch size of 32 to train the model and utilize the Adam optimization algorithm [29] with a learning rate of 1e-4, where the learning rate is selected from the set $lr =$ 1e-4, $lr\epsilon$ {1e-1, 1e-2, 1e-3, 1e-4}. The ReLU [30] activation function is employed to mitigate overfitting, and *dropout* regularization with a rate of 30% is applied, where the rate is selected from the set {20%, 30%, 50%} following the dense layers, using the grid search method. The output layers for the complaint classification, SC, ER, and SA tasks employ a softmax activation function with 2, 5, 7, and 3 neurons, respectively. In the CE task, the output layer features 20 neurons with a *Sigmoid* activation function. Our baseline model, SpanBERT, is sourced from the Hugging Face Transformer open-source repository[7]. Predictive performance is assessed by utilizing the accuracy mean and macro-F1 across ten nested 10-fold cross-validation experiments, with standard deviations also reported.

C. Resultant Findings With Discussion

This study aims to improve the performance of CD and SC when they are jointly learned with the CE task. The findings and the associated discussions are presented with CD, SC, and CE as the essential tasks in every possible combination to achieve this objective.

In this study, two distinct variants of a novel architecture have been introduced, namely the HAtt + CentralNet and WAtt + CentralNet multitask models. The rationale behind this is that not all elements of textual data are of equal importance in resolving a particular issue. It is imperative to model the relationships between words rather than their standalone presence to determine the critical components. The results of the experiments conducted on the CD, SC, and CE tasks are displayed in Table III, showcasing the effectiveness of the proposed architectures.

The results of the experiments indicate that the HAtt + CentralNetAll model exhibits superior performance over the baselines for both the CD and SC tasks. This superiority can be attributed to the ability of the model to effectively capture contextual information, which results in improved performance when compared to models that rely solely on WAtt mechanisms.

Furthermore, it can be deduced that the model HAtt + CentralNetCD + SC + CE + SA outperforms HAtt + CentralNetCD + SC + CE + ER. This can be attributed to the fact that the task of ER, being a highly context-dependent and fine-grained task, does not significantly contribute to the overall performance of the multitask model when solely considered.

The evaluation of the CE task was conducted using three distinct metrics, namely Jaccard similarity (JS), hamming distance (HD), and Ratcliff–Obershelp similarity (ROS). The primary purpose of these metrics is to quantitatively evaluate the CE task's effectiveness by analyzing token distance, edit distance, and sequence distance, respectively.

The ROS metric, which utilizes the longest sequence matching method, is deemed suitable for assessing the effectiveness of the common error (CE) task's training objective. The ROS metric yields a more precise indication of the model's performance. As indicated in Table III, the HAtt + $\text{CentralNet}_{\text{All}}$ model outperforms all the other models for the CE task, achieving a ROS score of 93.29. All of the results are statistically significant[8] [33].

1) Comparison With State-of-the-Art Technique (SOTA): We evaluate our proposed model to the current state-of-the-art model (SOTA_1) [6] for both CD and CS tasks. SOTA_1 employs a variety of pretrained transformer models having additional linguistic information. Both the proposed multitask models outperform the SOTA technique. Additionally, we also outperform the original financial CD work [11] (SOTA_2) for both the CD and SC tasks. Furthermore, we also evaluate our model's performance with SOTA_3 [31], a generative approach-based complaint identification model with SC configuration. The proposed HAtt + CentralNet model outperforms SOTA_3 for CD tasks by a significant margin. The possible reason for the better performance could be that the proposed model leverages dual attention mechanisms, leading to a better understanding of the context, also the addition of the CE task enhances the model's performance for the CD and SC tasks. This proves that the proposed architecture's efficient strategy

[4]https://pytorch.org/
[5]https://www.tensorflow.org/
[6]https://scikit-learn.org/stable/
[7]https://huggingface.co/model

[8]For the significance test, we used the paired t-test on the results and found that performance improvement in the proposed model over the state-of-the-art is significant (p-value < 0.05).

Authorized licensed use limited to: Sivas Bilim ve Teknoloji Universitesi. Downloaded on June 09,2026 at 07:36:48 UTC from IEEE Xplore. Restrictions apply.

TABLE III
RESULTS OF ALL THE EXPERIMENTS CONDUCTED FOR CD, SC, AND CE TASKS. F1: MACRO-F1 SCORE, A: ACCURACY. ALL THE METRICS ARE GIVEN IN %. BOLD-FACED VALUES INDICATE THE MAXIMUM SCORES OBTAINED. THE * INDICATES STATISTICALLY SIGNIFICANT RESULTS

	Complaint (CD)		Severity (SC)		Cause (CE)		
Model	**F1**	**A**	**F1**	**A**	**HD**	**JS**	**ROS**
SOTA_1 [6]	86.6	87.6	66.52	67.89	-	-	-
SOTA_2 [11]	85.82	86.33	64.83	66.23	-	-	-
WAtt + CentralNetCD + SC + CE	86.81	86.96	64.63	70.14	61.96	72.94	83.12
WAtt + CentralNetCD + SC + CE + SA	88.05	88.08	66.66	70.93	62.36	73.21	83.54
WAtt + CentralNetCD + SC + CE + ER	87.09	87.25	65.18	70.14	62.31	73.26	83.62
WAtt + CentralNetAll	88.11	88.12	66.93	71.22	62.76	73.54	83.73
HAtt + CentralNetCD + SC + CE	84.08	84.35	63.23	67.23	71.87	86.24	92.64
HAtt + CentralNetCD + SC + CE + SA	84.87	84.93	65.25	68.99	72.23	86.5	93.14
HAtt + CentralNetCD + SC + CE + ER	84.52	84.80	63.94	68.85	72.40	86.62	92.88
HAtt + CentralNetAll	**92.58***	**93.58***	**67.88***	**70.97***	**72.87**	**86.75**	**93.29**
SOTA_3 [31]	91.6	92.9	74.7	74.9	85.1	77.2	89.9
Baseline1 [5]	83.41	84.84	62.77	63.67	61.53	73.28	82.67
SpanBERT [32]	-	-	-	-	63.28	71.51	82.45

TABLE IV
QUALITATIVE ANALYSIS OF THE CD AND SC TASKS RESULTS BY THE PROPOSED (HATT + CENTRALNET) MODEL AND SOTA_1 MODEL. ''TRUE LABEL:'' ACTUAL LABELS FOR CD AND SC TASKS. THE BOLD-FACED TEXT REPRESENTS THE CAUSAL SPAN ANNOTATION OF THE INSTANCE

Tweet Text	**SOTA**	**Proposed**	**Actual Label**
@letsblinkit As per your msg, the **refund (of failed transaction on 23 December) date 28 December is gone,** you are not even replying neither updating, your official msg are not trustable,,,what kind do service is this...	Complaint Blame	Complaint Disapproval	Complaint Disapproval
@IDFCFIRST_Help **Stop giving the standard responses and tell me who authorized for the direct transaction** from my bank account @ICICIBank_Care@ICICIBank	Complaint Blame	Complaint No explicit reproach	Complaint No explicit reproach

is to use interactions between related tasks, especially between CD, SC, and CE.

a) Qualitative analysis: Tweets with dominant complaint cues, such as accusatory language or blame-indicating words, are correctly classified. Table IV shows the qualitative analysis of the best-performing HAtt + CentralNet model and SOTA_1 [6] for the severity prediction task on some sample test cases. It can be observed from the table that the proposed model attains refined predictions over the SOTA system that lacks the dyadic attention-based multitasking technique. One plausible explanation is the elimination of redundant information in the CE task, which results in prediction of more specific fine-grained severity levels. Table IV demonstrates that for both sample cases, both models accurately identify them as complaints, whereas only the proposed model precisely predicts the severity level. Evidently, from Table IV, the proposed model can handle causes of varied length. As shown in Table V, we also conducted a qualitative study for the CE task. The initial row depicts the causal span selected as significant for classification by human annotators. The following two rows illustrate the spans generated by the two proposed models. Both models extract correct causal spans from the given text with minor variations from the human annotator-picked span.

b) Error analysis: We investigated some of the reasons for the proposed model's misclassification.

1) *Dispersed causes:* The model cannot identify noncontinuous causes scattered over a tweet instance. For example, @HDFCBank_*Cares Credit card blocked post a valid transaction. No callback. Not able to reach customer care amidst plethora of IVR options. Please help.* The causal span predicted: *Credit card blocked post a valid transaction.* The present study has identified the causal factors through the process of annotation, where the first interaction that contains a clear and specific declaration

Authorized licensed use limited to: Sivas Bilim ve Teknoloji Universitesi. Downloaded on June 09,2026 at 07:36:48 UTC from IEEE Xplore. Restrictions apply.

TABLE V
EXAMPLES OF CAUSES INDICATED BY HUMAN ANNOTATORS AND SPANBERT BASELINE AGAINST THE HATT+CENTRALNET$_{ALL}$ AND WATT + CENTRALNET$_{ALL}$ MODELS. THE HUMAN ANNOTATOR AND THE PROPOSED MODELS' SELECTED CAUSAL SPANS ARE INDICATED IN RED COLORED TEXT. TOKENS THAT ARE HIGHLIGHTED IN BLUE MAY NOT HOLD ANY SIGNIFICANCE FOR HUMAN ANNOTATORS. HOWEVER, THE PROPOSED MODEL CONSIDERS THESE TOKENS TO BE RELEVANT. (THIS IS BEST VIEWED IN COLOR)

Model	Text
Human Annotator	< USER > Please help me to settle my loan request its already 1 month passed.
SpanBERT [32]	< USER > Please help me to settle my loan request its already 1 month passed.
HAtt + CentralNet$_{All}$	< USER > Please help me to settle my loan request its already 1 month passed.
WAtt + CentralNet$_{All}$	< USER > Please help me to settle my loan request its already 1 month passed.

regarding the cause of the complaint is taken into account. The proposed model cannot identify causes distributed (for instance, *No callback. Not able to reach customer care*) throughout the entire tweet.

2) *Misclassified severity:* Instances from adjoining severity levels are often conceptually related, resulting in misinterpretation. In the given example, *Fraud transaction happened in my bank account, did not receive any explanation. Highly disappointed with the response and lived in fear of losing my hard-earned money.* The model prediction is disapproval of the severity level, even though the true level for the current example is an accusation. As a result of the presence of negative connotations such as *disappointed*, which are often found in the disapproval class, the proposed model incorrectly categorizes it.

VI. LIMITATIONS

The proposed framework has some textual shortcomings mentioned as follows

1) *Handling complexity:* The model may encounter difficulties in processing complex and contextually nuanced situations. This could lead to potential inaccuracies and limitations in capturing the full scope of complaints.
2) *Limited causal span identification:* The present approach relies on identifying causal factors through specific declarations in the initial interactions. This may miss important causes distributed throughout the entire tweet, reducing the model's ability to grasp the full context.

Considering these limitations, further research and refinements are necessary to enhance the model's capabilities in addressing the diverse and intricate language patterns prevalent in financial complaints on social media.

VII. CONCLUSION

Our research introduced a new objective called rationale extraction which aims to extract reasoning from a collection of financial complaints using a multitask approach that utilizes HAtt. In order to advance research on the explainable identification of complaints, an extension has been made to the *FINCORP* dataset by incorporating a gold standard annotation of the cause for each individual instance. Based on the findings of this study, we observed that our proposed HAtt + CentralNet$_{All}$ model outperformed SOTA by capturing contextual information with 93.58% accuracy. Moreover, the CE task improved performance by accurately identifying the longest sequence of causes. This shows that complaint identification is more than just classifying data—it's about preventing expectation violations. Conclusively, our proposed method's interpretability is a significant advantage in evaluating the automated system's influence and prediction confidence.

In future work, the emphasis will be on enhancing the ability to identify multiple causes and the utilization of multiple sources of information in the detection and extraction of causal spans, with the objective of increasing the interpretability of the cause analysis method that has been put forward.

ACKNOWLEDGMENT

The authors express their gratitude for the assistance provided by CRISIL Data Science Private Ltd. and the team of dataset annotators, specifically Rik Biswas and Aakansha Prasad, for their unwavering dedication and contributions.

REFERENCES

[1] E. Olshtain and L. Weinbach, "Complaints: A study of speech act behavior among native and nonnative speakers of Hebrew," in *The pragmatic perspective*, John Benjamins, 1987, pp. 1–195.
[2] A. Trosborg, *Interlanguage Pragmatics: Requests, Complaints, and Apologies*, vol. 7. Berlin, Germany: Walter de Gruyter, 2011.
[3] L. N. Kakolaki and M. Shahrokhi, "Gender differences in complaint strategies among Iranian upper intermediate EFL students," *Stud. English Lang. Teach.*, vol. 4, no. 1, pp. 1–15, 2016.
[4] D. Preotiuc-Pietro, M. Gaman, and N. Aletras, "Automatically identifying complaints in social media," in *Proc. 57th Conf. Assoc. Comput. Linguistics (ACL)*, Florence, Italy, A. Korhonen, D. R. Traum, and L. Màrquez, Eds., vol. 1. Association for Computational Linguistics, Jul. 28–Aug. 2, 2019, pp. 5008–5019.
[5] A. Singh, A. Nazir, and S. Saha, "Adversarial multi-task model for emotion, sentiment, and sarcasm aided complaint detection," in *Proc. Adv. Inf. Retrieval 44th Eur. Conf. IR Res. (ECIR) I, Lecture Notes Comput. Sci.*, Stavanger, Norway, M. Hagen et al., Eds., vol. 13185. Berlin, Germany: Springer, Apr. 10–14, 2022, pp. 428–442, doi: 10.1007/978-3-030-99736-6_29.
[6] M. Jin and N. Aletras, "Modeling the severity of complaints in social media," in *Proc. Conf. North Amer. Chapter Assoc. Comput. Linguistics, Hum. Lang. Technol. (NAACL-HLT)*, K. Toutanova et al., Eds. Association for Computational Linguistics, Jun. 6–11, 2021, pp. 2264–2274, doi: 10.18653/v1/2021.naacl-main.180.
[7] D. Gunning, M. Stefik, J. Choi, T. Miller, S. Stumpf, and G. Yang, "XAI - Explainable artificial intelligence," *Sci. Robot.*, vol. 4, no. 37, 2019, doi: 10.1126/scirobotics.aay7120.

Authorized licensed use limited to: Sivas Bilim ve Teknoloji Universitesi. Downloaded on June 09,2026 at 07:36:48 UTC from IEEE Xplore. Restrictions apply.

[8] N. D. Marcellis-Warin, W. Sanger, and T. Warin, "A network analysis of financial conversations on Twitter," *Int. J. Web Based Communities*, vol. 13, no. 3, pp. 281–310, 2017.
[9] S. Y. M. Lee, Y. Chen, and C.-R. Huang, "A text-driven rule-based system for emotion cause detection," in *Proc. NAACL HLT Workshop Comput. Approaches Anal. Gener. Emotion Text*, 2010, pp. 45–53.
[10] S. Poria et al., "Recognizing emotion cause in conversations," *Cogn. Comput.*, vol. 13, no. 5, pp. 1317–1332, 2021.
[11] A. Singh, R. Bhatia, and S. Saha, "Complaint and severity identification from online financial content," *IEEE Trans. Comput. Social Syst.*, early access, 2023.
[12] K. Coussement and D. Van den Poel, "Improving customer complaint management by automatic email classification using linguistic style features as predictors," *Decis. Support Syst.*, vol. 44, no. 4, pp. 870–882, 2008.
[13] M. Jin and N. Aletras, "Complaint identification in social media with transformer networks," in *Proc. 28th Int. Conf. Comput. Linguistics (COLING)*, Barcelona, Spain (Online), D. Scott, N. Bel, and C. Zong, Eds., Dec. 8–13, 2020, pp. 1765–1771, doi: 10.18653/v1/2020.coling-main.157.
[14] A. Singh, V. Gangwar, S. Sharma, and S. Saha, "Knowing what and how: A multi-modal aspect-based framework for complaint detection," in *Proc. Adv. Inf. Retrieval, 45th Eur. Conf. Inf. Retrieval II (ECIR)*, Dublin, Ireland. Springer, Apr. 2–6, 2023, pp. 125–140.
[15] S. Bhat and A. Culotta, "Identifying leading indicators of product recalls from online reviews using positive unlabeled learning and domain adaptation," in *Proc. Int. AAAI Conf. Web Social Media*, 2017, vol. 11.
[16] W. Yang et al., "Detecting customer complaint escalation with recurrent neural networks and manually-engineered features," in *Proc. Conf. North Amer. Chapter Assoc. Comput. Linguistics, Hum. Lang. Technol. (NAACL-HLT)*, Minneapolis, MN, USA, A. Loukina, M. Morales, and R. Kumar, Eds., vol. 2. Association for Computational Linguistics, Jun. 2–7, 2019, pp. 56–63, doi: 10.18653/v1/n19-2008.
[17] M. Isonuma, T. Fujino, J. Mori, Y. Matsuo, and I. Sakata, "Extractive summarization using multi-task learning with document classification," in *Proc. Conf. Empirical Methods Natural Lang. Process.*, 2017, pp. 2101–2110.
[18] L. Yang, T. L. J. Ng, B. Smyth, and R. Dong, "HTML: Hierarchical transformer-based multi-task learning for volatility prediction," in *Proc. Web Conf.*, 2020, pp. 441–451.
[19] R. Xia and Z. Ding, "Emotion-cause pair extraction: A new task to emotion analysis in texts," 2019, *arXiv:1906.01267*.
[20] P. Ekman et al., "Universals and cultural differences in the judgments of facial expressions of emotion," *J. Personality Social Psychol.*, vol. 53, no. 4, p. 712, 1987.
[21] P. Rajpurkar, J. Zhang, K. Lopyrev, and P. Liang, "Squad: 100,000+ questions for machine comprehension of text," 2016, *arXiv:1606.05250*.
[22] J. Devlin, M. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of deep bidirectional transformers for language understanding," in *Proc. Conf. North Amer. Chapter Assoc. Comput. Linguistics, Hum. Lang. Technol. (NAACL-HLT)*, Minneapolis, MN, USA, J. Burstein, C. Doran, and T. Solorio, Eds., vol. 1. Association for Computational Linguistics, Jun. 2–7, 2019, pp. 4171–4186, doi: 10.18653/v1/n19-1423, *arXiv:1810.04805*.
[23] A. Vaswani et al., "Attention is all you need," 2017, *arXiv:1706.03762*.
[24] Z. Liu, W. Lin, Y. Shi, and J. Zhao, "A robustly optimized BERT pre-training approach with post-training," in *China National Conference on Chinese Computational Linguistics*, Springer, 2021, pp. 471–484.
[25] Y. Liu, H. Liu, L.-P. Wong, L.-K. Lee, H. Zhang, and T. Hao, "A hybrid neural network rBERT-c based on pre-trained ROBERTa and CNN for user intent classification," in *Neural Comput. Adv. Appl.: First Int. Conf. (NCAA)*, Shenzhen, China, Springer, Jul. 3–5, 2020, pp. 306–319.
[26] V. Vielzeuf, A. Lechervy, S. Pateux, and F. Jurie, "CentralNet: A multilayer approach for multimodal fusion," in *Proc. Eur. Conf. Comput. Vision (ECCV) Workshops*, 2018.
[27] A. Paszke et al., "Automatic differentiation in PyTorch," 2017.
[28] F. Pedregosa et al., "Scikit-learn: Machine learning in python," *J. Mach. Learn. Res.*, vol. 12, pp. 2825–2830, Oct. 2011.
[29] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," 2014, *arXiv:1412.6980*.
[30] X. Glorot, A. Bordes, and Y. Bengio, "Deep sparse rectifier neural networks," in *Proc. 14th Int. Conf. Artif. Intell. Statist.*, 2011, pp. 315–323.
[31] S. Das, A. Singh, R. Jain, S. Saha, and A. Maurya, "Let the model make financial senses: A text2text generative approach for financial complaint identification," in *Proc. Pacific-Asia Conf. Knowl. Discovery Data Mining*, Berlin, Germany: Springer, 2023, pp. 58–69.
[32] M. Joshi, D. Chen, Y. Liu, D. S. Weld, L. Zettlemoyer, and O. Levy, "SpanBERT: Improving pre-training by representing and predicting spans," *Trans. Assoc. Comput. Linguistics*, vol. 8, pp. 64–77, 2020.
[33] B. L. Welch, "The generalization of 'student's' problem when several different population varlances are involved," *Biometrika*, vol. 34, nos. 1–2, pp. 28–35, 1947.

Authorized licensed use limited to: Sivas Bilim ve Teknoloji Universitesi. Downloaded on June 09,2026 at 07:36:48 UTC from IEEE Xplore. Restrictions apply.